Network-Based Simulation of COVID-19 Transmission Dynamics in Colombia Using the SIR Model

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This project presents a graph-based simulation framework for modeling the spread of COVID-19 in Colombia using real case data from the National Institute of Health (INS). Confirmed cases were represented as nodes in an undirected weighted contagion graph, while edges encoded possible epidemiological relationships based on geographic and temporal proximity. A discrete SIR model was implemented over this network, allowing disease transmission to occur locally through graph edges rather than assuming homogeneous mixing across the population.
The constructed network contains 1,161 nodes and 377 edges, showing a highly sparse and fragmented structure with 784 connected components. Several graph algorithms were incorporated to analyze the network, including BFS for contact tracing, Dijkstra's algorithm for probable contagion paths, and centrality measures for identifying influential nodes. Four containment scenarios were evaluated: no intervention, hub isolation, BFS-based contact tracing, and targeted vaccination using betweenness centrality.
The results show that hub isolation was the most effective strategy, reducing the infection peak from 540 to 435 infected nodes, corresponding to a 19.4% reduction relative to the baseline. Contact tracing achieved an intermediate reduction, while betweenness-based vaccination produced only a marginal effect due to the fragmented topology of the network. Additional simulations on Erdős–Rényi, Watts–Strogatz, and Barabási–Albert synthetic networks confirmed that epidemic dynamics depend not only on SIR parameters, but also on network topology and initial infection conditions. Overall, this work demonstrates how graph theory and classical epidemic modeling can be combined to study contagion dynamics and evaluate containment strategies from a computational perspective.

1. Introduction

The COVID-19 pandemic emphasized the need for computational models to understand how infectious diseases spread across populations. Although classical epidemiological models such as SIR describe the evolution of susceptible, infected, and recovered individuals, they often simplify the contact structure of a real population. Graphs provide a useful alternative, since they allow cases and possible transmission links to be represented explicitly.
In this project, COVID-19 cases in Colombia are modeled as an undirected weighted graph, where each node represents a confirmed case and each edge represents a possible epidemiological relationship based on department and temporal proximity. Over this graph, an SIR simulation is implemented to analyze the propagation process and compare different containment strategies.
Additionally, graph algorithms such as BFS, DFS, Dijkstra, degree centrality, and betweenness centrality are used to study contact paths, connected components, probable contagion chains, and influential nodes. The real contagion network is also compared with synthetic networks, including Erdős–Rényi, Watts–Strogatz, and Barabási-Albert models, to evaluate how network topology affects disease spread.

SIR Model for Epidemiology

The Susceptible–Infected–Recovered (SIR) model provides a mathematical framework for describing the temporal evolution of an epidemic process. In its classical formulation, a population of size N is divided into three disjoint compartments: susceptible individuals $S(t)$, infected individuals $I(t)$, and recovered individuals $R(t)$. These quantities satisfy

$$S(t) + I(t) + R(t) = N,$$

for every time t. The basic transition mechanism of the model is

$$S \longrightarrow I \longrightarrow R,$$

where susceptible individuals become infected through contact with infected individuals, and infected individuals eventually recover.

In the classical continuous formulation, the dynamics are commonly expressed by the system of differential equations

$$\frac{dS}{dt} = -\beta \frac{SI}{N},$$
$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I,$$
$$\frac{dR}{dt} = \gamma I,$$

where β denotes the transmission rate and γ denotes the recovery rate. The term $\beta SI/N$ models the rate at which susceptible individuals become infected, while γI represents the rate at which infected individuals recover. Although this formulation is useful, it assumes homogeneous mixing, meaning that every individual is equally likely to interact with every other individual in the population.

In this project, the SIR process is adapted to a graph-based setting. Let

$$G = (V, E)$$

be an undirected weighted graph, where each vertex $v \in V$ represents a confirmed COVID-19 case and each edge $(u, v) \in E$ represents a possible epidemiological relationship between two cases. The edge weights encode the strength or likelihood of a connection between cases, based on criteria such as geographic proximity and temporal closeness. Thus, disease transmission is not modeled as a global interaction among all individuals, but as a local process determined by the neighborhood structure of the graph.

At each discrete time step t, every node has a state

$$X_v(t) \in \{S, I, R\},$$

where $X_v(t) = S$ means that node v is susceptible, $X_v(t) = I$ means that it is infected, and $X_v(t) = R$ means that it has recovered. A susceptible node can become infected only if it is adjacent to at least one infected node. Therefore, the infection probability of a node depends on its infected neighborhood,

$$\mathcal{N}_I(v, t) = \{u \in \mathcal{N}(v) : X_u(t) = I\}.$$

This makes the epidemic evolution strongly dependent on the topology of the graph.

A natural way to express the infection rule is

$$\mathbb{P}\big(X_v(t+1) = I \mid X_v(t) = S\big) = 1 - \prod_{u \in \mathcal{N}_I(v,t)} (1 - \beta w_{uv}),$$

where w_{uv} is the weight of the edge connecting nodes u and v. This expression represents the probability that at least one infected neighbor transmits the disease to node v. If a susceptible node has no infected neighbors, then its probability of becoming infected is zero.

Recovery is modeled by a transition from the infected state to the recovered state. In probabilistic form, this can be written as

$$\mathbb{P}\big(X_v(t+1) = R \mid X_v(t) = I\big) = \gamma,$$

where γ is the recovery probability per time step. Once a node reaches the recovered state, it no longer participates in the transmission process. Hence, recovered nodes are treated as inactive with respect to future infections.

This graph-based formulation is more suitable for contagion analysis than the homogeneous SIR model when the underlying contact structure is relevant. In particular, nodes with high degree may accelerate the spread because they have many neighbors, while nodes with high betweenness centrality may connect otherwise separated regions of the network. As a result, the epidemic curve is influenced not only by the parameters β and γ, but also by structural properties of the graph such as degree distribution, connected components, shortest paths, and centrality measures.

Graph Sfor Network Analysis

In addition to the SIR epidemic dynamics, several graph algorithms were implemented to characterize the structure of the contagion network. These algorithms provide complementary information about connectivity, possible transmission paths, and influential nodes within the graph.

Breadth-First Search (BFS) was used to explore the network by levels and to support contact-tracing analysis. Given an initial infected node, BFS allows the identification of nodes that can be reached within a fixed number of steps, which can be interpreted as possible chains of indirect exposure.

Depth-First Search (DFS) was used to identify connected components in the graph. This is relevant because disease propagation is constrained by connectivity: if two nodes belong to different connected components, there is no transmission path between them under the constructed network structure.

Dijkstra's algorithm was implemented to estimate the most likely contagion path between two cases. Since the graph is weighted, shortest paths can be interpreted as paths with stronger epidemiological proximity, depending on how edge weights are defined.

Finally, degree centrality and betweenness centrality were used to identify structurally important nodes. Degree centrality highlights highly connected nodes that may act as local spreaders, while betweenness centrality identifies nodes that serve as bridges between different regions of the network. These measures are especially useful when evaluating containment strategies, since isolating central nodes may reduce the speed or extent of propagation.

Methodology

The simulator was developed using real COVID-19 case data from the Colombian National Institute of Health (INS). In particular, the file `Casos1.csv`, corresponding to the first wave of the pandemic in Colombia during March–April 2020, was used as the main dataset. This dataset contains 1,161 individual confirmed cases, including attributes such as case identifier, diagnosis date, department, city, age, sex, and case type. Additionally, the file `covid-19-colombia.csv`, containing daily national totals, was used to calibrate the epidemiological parameters of the model.

A. Graph Construction. The contagion network was modeled as an undirected weighted graph

$$G = (V, E),$$

where each node $v \in V$ represents a confirmed COVID-19 case and each edge $(u, v) \in E$ represents a possible epidemiological relationship between two cases. Since the resulting graph is sparse, with 1,161 nodes and 377 edges, an adjacency-list representation was selected instead of an adjacency matrix. This structure was implemented as

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which provides efficient memory usage of order $O(|V| + |E|)$ and allows neighborhood traversal proportional to the degree of each node.

Edges were constructed using the case type reported in the dataset. Cases classified as *Imported* were considered initial infected nodes, while cases classified as *Related* were connected to the temporally closest imported case within the same department, using an epidemiological window of 14 days. The weight of each edge was defined as

$$w_{uv} = \frac{1}{1 + \Delta t},$$

where Δt is the absolute difference in days between the diagnosis dates of the two cases. Under this definition, cases that are closer in time receive larger edge weights, representing stronger epidemiological proximity and a higher potential likelihood of transmission.

B. SIR Dynamics on the Graph. The classical Susceptible–Infected–Recovered (SIR) model was adapted to the graph structure. Instead of assuming homogeneous mixing in the whole population, the graph-based version assigns an individual epidemiological state to each node and restricts transmission to adjacent nodes. Thus, each node v has a state

$$X_v(t) \in \{S, I, R\},$$

where S denotes susceptible, I infected, and R recovered.

At each discrete simulation step, the following transition rules were applied. First, each infected node attempts to infect each susceptible neighbor with probability $P_{u \to v} = \beta w_{uv}$ where β is the transmission rate and w_{uv} is the weight of the edge connecting the two nodes. Second, each infected node recovers with probability γ. Therefore, the epidemic evolves locally according to the connectivity structure of the graph.

The parameters were calibrated from the national COVID-19 time series. The transmission rate was set to $\beta = 0.32$ and the recovery rate to $\gamma = 0.071$, which corresponds approximately to an infectious period of 14 days. These values yield a basic reproductive number

$$R_0 = \frac{\beta}{\gamma} \approx 4.5.$$

Although this quantity comes from the classical SIR interpretation, in the graph-based model the effective spread also depends on structural properties such as degree distribution, connected components, and centrality.

C. Containment Strategies. Four containment strategies were implemented and compared.

The first scenario corresponds to a baseline simulation without intervention. In this case, the graph structure remains unchanged and the SIR process evolves naturally according to the transmission and recovery probabilities.

The second strategy consists of isolating hubs. Degree centrality was computed for all nodes, and the top 5% of nodes with the largest number of connections were removed from the graph. This strategy is motivated by the fact that highly connected nodes can accelerate disease propagation by exposing many neighboring nodes.

The third strategy simulates contact tracing using Breadth-First Search (BFS). Starting from the initially infected nodes, BFS was used to identify direct contacts at distance one. The edges associated with these contacts were removed, representing an immediate quarantine mechanism that interrupts local transmission routes.

The fourth strategy corresponds to targeted vaccination based on betweenness centrality. Betweenness centrality was computed using Brandes' algorithm, and the top 10% of nodes with the highest values were directly assigned to the recovered state. This intervention targets nodes that act as bridges between different parts of the graph, with the goal of reducing inter-component transmission.

D. Synthetic Network Models. To evaluate how network topology affects epidemic propagation, three synthetic network models were also generated. The Erdős–Rényi model $G(n, p)$ was used to represent random connectivity, the Watts–Strogatz model was used to generate small-world networks, and the Barabási–Albert model was used to represent scale-free networks with preferential attachment. These synthetic networks provide a reference framework for comparing the behavior of the real contagion graph against standard network structures.

E. Implementation. The simulator was implemented in C++17 and organized into independent modules. The file `graph.cpp` contains the graph structure and traversal algorithms, `model.cpp` implements the SIR simulation and containment strategies, `data_upd.cpp` handles CSV loading and preprocessing, `dijkstra.cpp` performs shortest-path analysis, and `network_gen.cpp` generates synthetic networks. The simulation outputs are exported as CSV files and visualized in Python using `matplotlib`.

Results

Network Structure and Topology. The contagion graph constructed from the INS dataset contains 1,161 nodes and 377 edges, with a density of 0.0006. These values confirm that the resulting network is highly sparse. Figure 1 shows both the full graph and its giant connected component. The graph is divided into 784 connected components, while the largest component contains only 23 nodes. This fragmentation is consistent with the epidemiological context of the first COVID-19 wave in Colombia during March–April 2020, when imported cases and early isolation measures limited the formation of large transmission clusters.

The mean degree of the network is 0.65, whereas the maximum degree is 22. The node with the highest degree corresponds to case ID 1030, an imported case that behaves as the dominant local hub in the constructed graph. This indicates that, although most nodes have very few connections, a small number of cases concentrate a relatively large number of possible epidemiological links. Such degree heterogeneity is relevant because it directly affects the performance of targeted containment strategies.

Table 1. Structural metrics of the contagion network.

Metric	Value
Nodes	1,161
Edges	377
Density	0.0006
Mean degree	0.65
Maximum degree	22
Connected components	784
Giant component	23 nodes

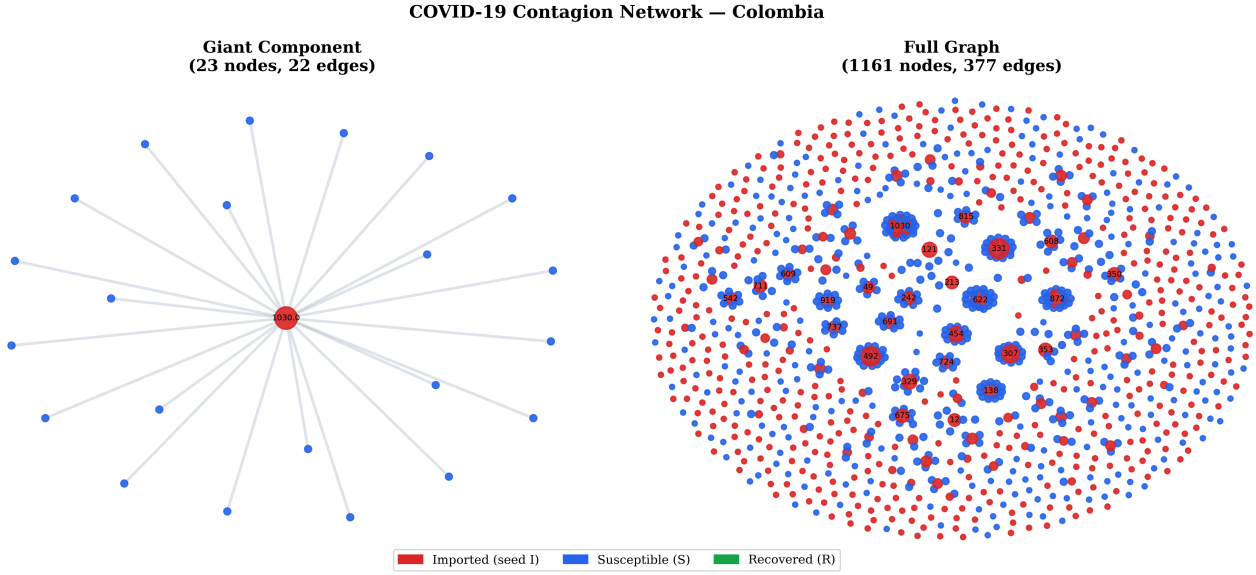


Fig. 1. COVID-19 contagion network constructed from INS Colombia data. Left: giant connected component. Right: full graph with 1,161 nodes and 377 edges. Node color indicates initial SIR state, and edge width is proportional to the transmission weight $w_{uv} = 1/(1 + \Delta t_{uv})$.

Degree Distribution and Centrality. Figure 2 presents the degree distribution of the contagion graph, the degree distribution by case type, and the ranking of departments by number of cases. The degree distribution is strongly right-skewed: most nodes

have degree zero or one, while a small number of nodes reach much higher degree values. This pattern suggests heterogeneous connectivity, where a few cases may act as local spreaders within the network.

Imported cases tend to have higher degree values than other case types. This is expected from the graph construction procedure, since imported cases were used as initial sources of infection and related cases were connected to temporally close imported cases within the same department. At the territorial level, Bogotá D.C. and Antioquia account for the largest number of confirmed cases, which is consistent with their demographic weight and their role as major mobility and transportation centers.

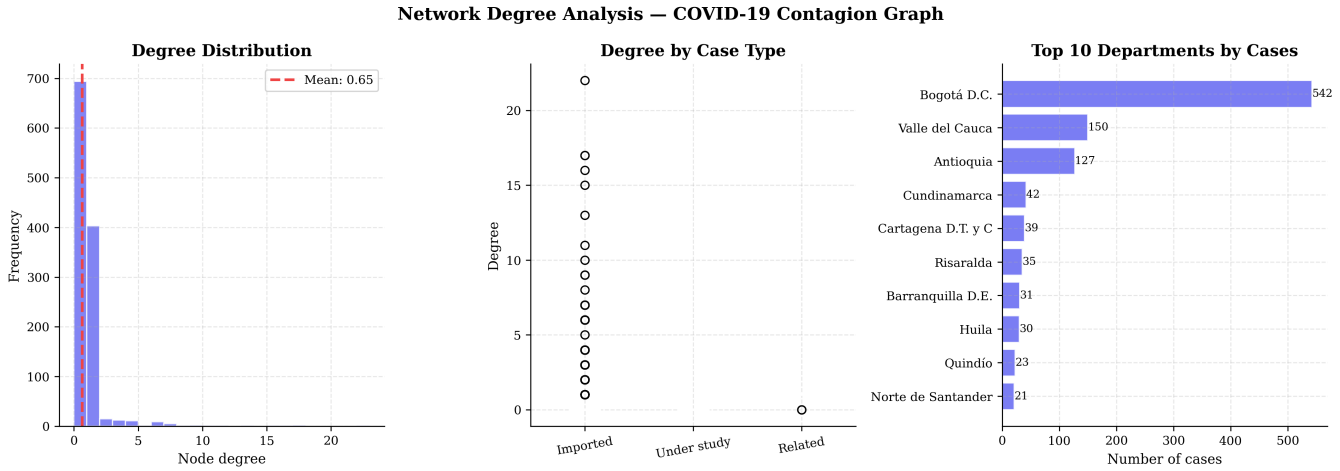


Fig. 2. Degree analysis of the contagion network. Left: degree distribution. Center: degree distribution by case type. Right: top 10 departments by confirmed case count.

Figure 3 compares degree centrality $C_D(v)$ and betweenness centrality $C_B(v)$ for all nodes. Case ID 1030 stands out under both measures, confirming its role as both a highly connected node and a structurally important connector within the largest transmission chain. However, the relation between degree and betweenness is not perfectly linear. Some nodes with moderate degree may still exhibit relatively high betweenness, indicating that they connect otherwise weakly connected parts of the graph. This distinction motivates the comparison between degree-based hub isolation and betweenness-based targeted vaccination.

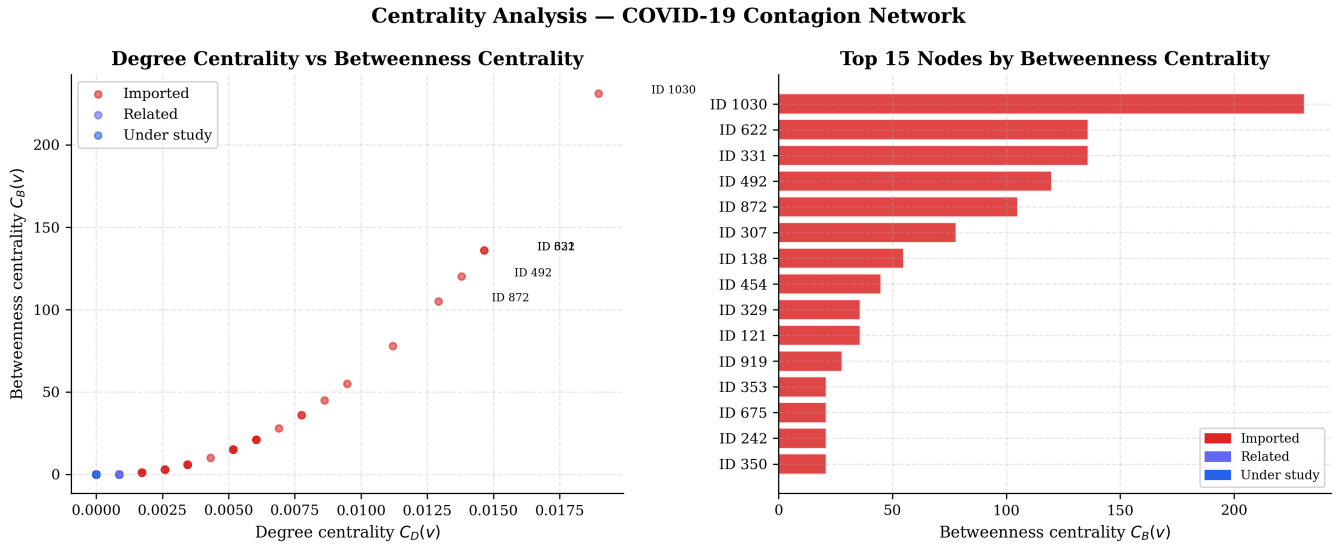


Fig. 3. Centrality analysis of the contagion network. Left: degree centrality versus betweenness centrality. Right: top nodes ranked by betweenness centrality $C_B(v)$.

SIR Epidemic Curves by Containment Strategy. Figures 4–7 show the SIR epidemic curves obtained under each containment strategy. These curves allow the temporal behavior of susceptible, infected, and recovered nodes to be compared under different interventions.

In the no-intervention scenario, shown in Figure 4, the epidemic reaches its maximum on day 2 with 540 infected nodes. The early peak is mainly explained by the large number of initially infected nodes in the real dataset. After the peak, the

infected population decreases as nodes transition to the recovered state. By day 60, 757 nodes have reached the recovered compartment, representing the cumulative epidemic burden under the baseline scenario.

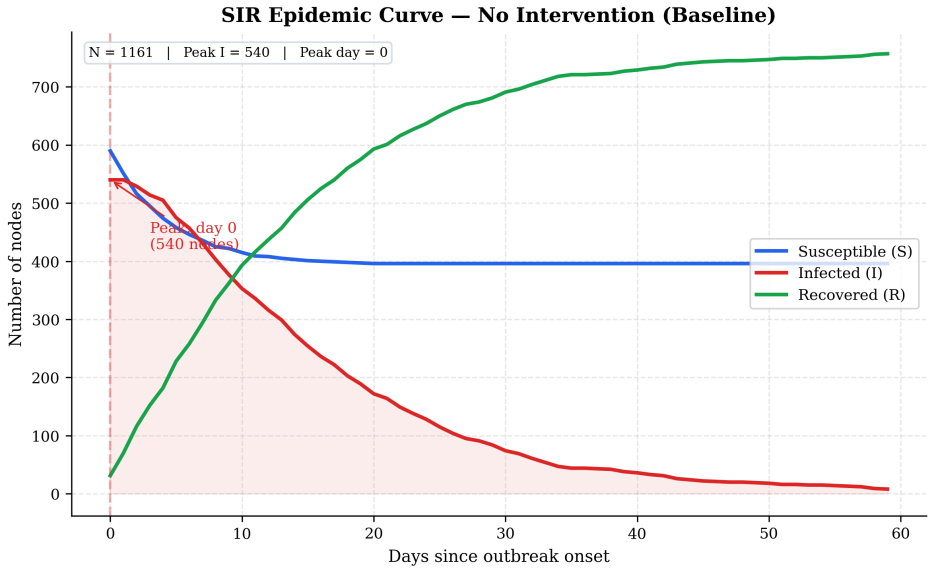


Fig. 4. SIR epidemic curve under no intervention. The infection peak occurs on day 2 with 540 infected nodes.

Hub isolation, shown in Figure 5, removes the top 5% of nodes by degree before the simulation begins. This intervention eliminates 58 highly connected nodes, including the dominant hub ID 1030. As a result, the infection peak decreases to 435 infected nodes, corresponding to a reduction of 19.4% relative to the baseline. The final number of recovered nodes also decreases to 490, indicating a substantial reduction in cumulative incidence. This result suggests that degree-based interventions are particularly effective when transmission is locally amplified by highly connected nodes.

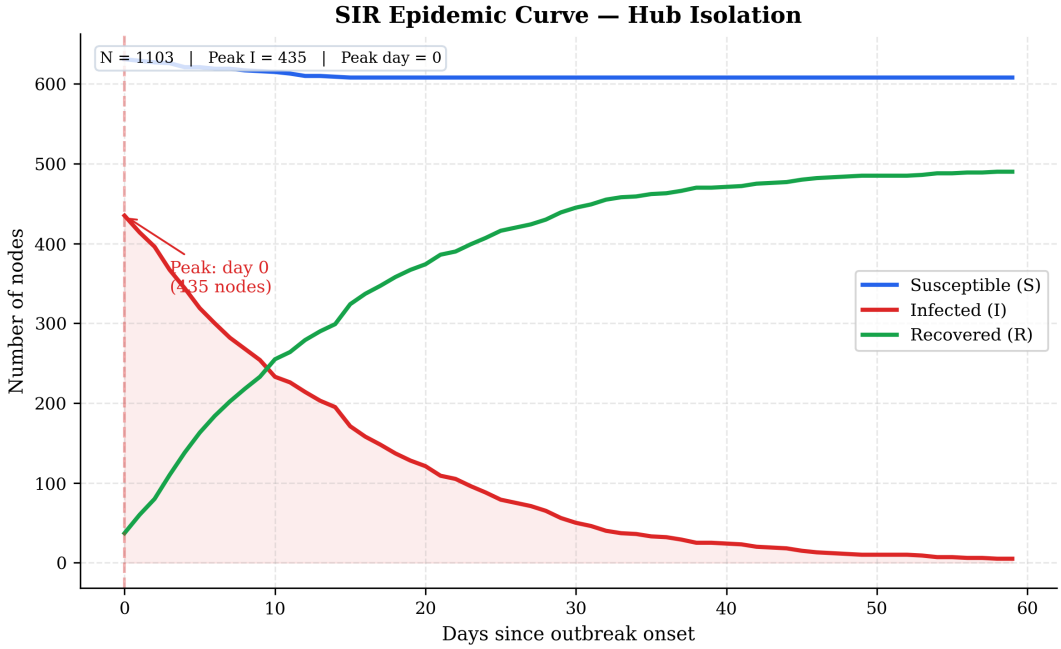


Fig. 5. SIR epidemic curve under hub isolation. The top 5% of nodes by degree centrality are removed before the simulation, reducing the infection peak and cumulative incidence.

Contact tracing via BFS, shown in Figure 6, performs a one-hop breadth-first search from the initially infected nodes and removes the corresponding contact edges. This strategy reduces the infection peak to 491 infected nodes, corresponding to a

reduction of 9.1% relative to the baseline. Although less effective than hub isolation, this strategy has a practical advantage: it relies on local information and interrupts specific contact links rather than removing individuals from the network.

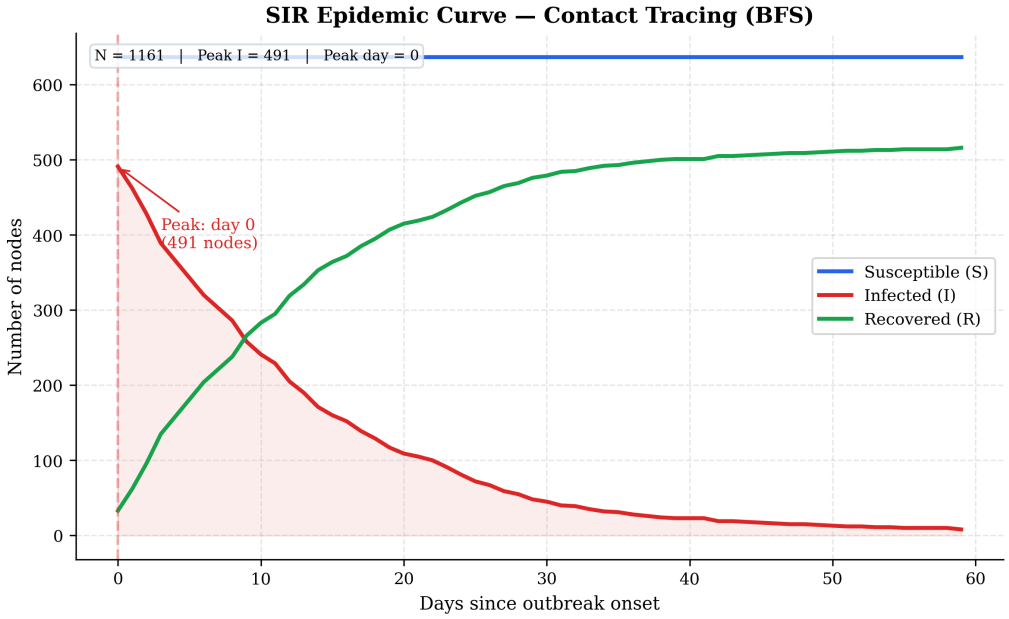


Fig. 6. SIR epidemic curve under BFS-based contact tracing. One-hop contacts from initially infected nodes are quarantined through edge removal.

Targeted vaccination based on betweenness centrality, shown in Figure 7, assigns the top 10% of nodes by $C_B(v)$ directly to the recovered state before the simulation. This strategy produces only a marginal reduction, with a peak of 523 infected nodes, equivalent to a 3.1% reduction relative to the baseline. The limited effect can be explained by the fragmented structure of the graph: in a network with many small components, nodes with high betweenness may act as local bridges without necessarily controlling the main transmission routes.

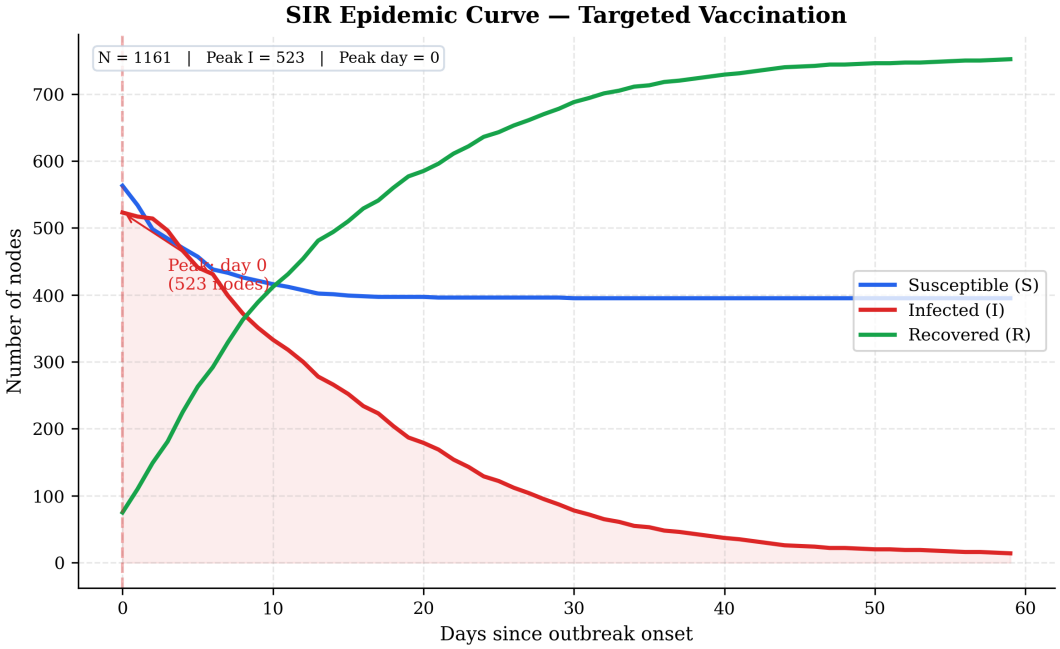


Fig. 7. SIR epidemic curve under targeted vaccination by betweenness centrality. The top nodes according to $C_B(v)$ are assigned to the recovered state before the simulation.

Comparative Analysis of Containment Strategies. Figure 8 summarizes the performance of the four containment strategies. Hub isolation achieves the largest reduction in both the infection peak and the final recovered population. The infection

peak decreases from 540 infected nodes in the baseline scenario to 435 under hub isolation, while the final recovered count decreases from 757 to 490. This confirms that removing highly connected nodes substantially reduces the epidemic burden in the constructed graph.

Contact tracing via BFS provides an intermediate reduction. It lowers the peak to 491 infected nodes and reduces the final recovered population to 516. Although its effect is smaller than that of hub isolation, its computational simplicity and local nature make it a realistic intervention mechanism. In contrast, targeted vaccination by betweenness centrality produces the weakest reduction among the three intervention strategies. This result suggests that betweenness centrality is not always the most informative criterion in sparse and disconnected networks. In this graph, degree centrality provides a more direct indicator of local spreading potential.

Table 2. Simulation results under different containment strategies over 60 time steps.

Strategy	Peak infected	Peak day	Reduction vs. baseline
No intervention	540	Day 2	—
Hub isolation	435	Day 0	-19.4%
Contact tracing (BFS)	491	Day 0	-9.1%
Targeted vaccination	523	Day 1	-3.1%

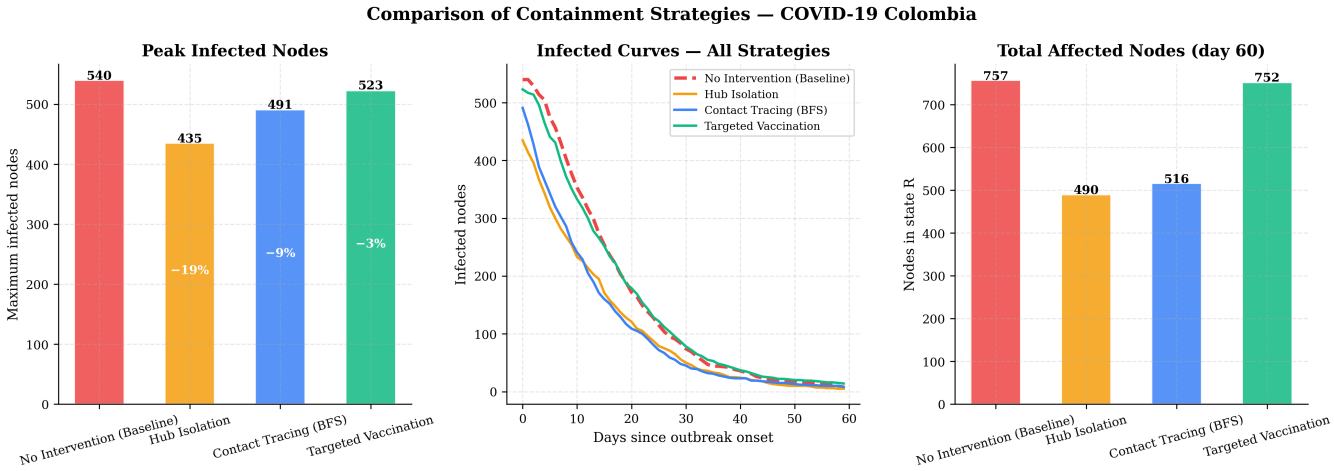


Fig. 8. Comparative analysis of containment strategies. Left: peak infected nodes per strategy. Center: temporal evolution of infected nodes. Right: total nodes in state R at day 60, interpreted as cumulative epidemic burden.

Contagion Path Analysis via Dijkstra's Algorithm. Figure 9 shows the most probable contagion chain obtained with Dijkstra's algorithm inside the giant component. Since the original edge weights represent epidemiological proximity, the transformation

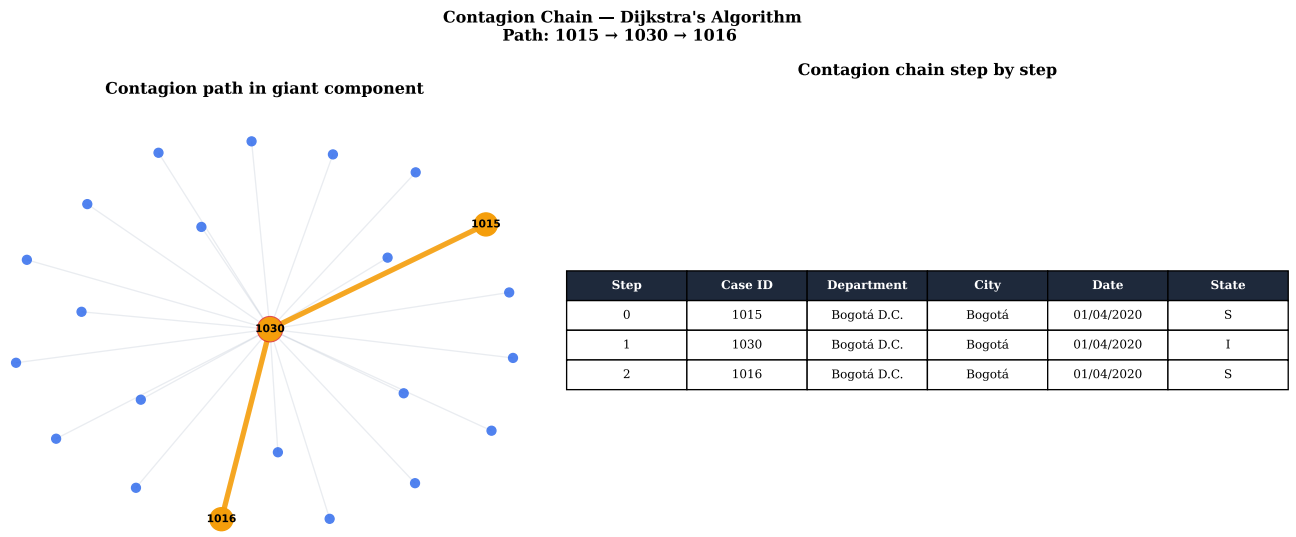
$$w_{uv} \mapsto -\log(w_{uv})$$

was used to convert the problem of maximizing accumulated transmission likelihood into a shortest-path problem.

The recovered path follows the sequence

$$1015 \to 1030 \to 1016.$$

The presence of node 1030 as an intermediate node is consistent with its role as the main hub of the network. This result illustrates how shortest-path algorithms can be used to reconstruct plausible transmission chains and support retrospective outbreak analysis.



Step	Case ID	Department	City	Date	State
0	1015	Bogotá D.C.	Bogotá	01/04/2020	S
1	1030	Bogotá D.C.	Bogotá	01/04/2020	I
2	1016	Bogotá D.C.	Bogotá	01/04/2020	S

Fig. 9. Most probable contagion chain identified by Dijkstra's algorithm. The highlighted path passes through node 1030, the dominant hub in the giant component.

Comparison with Synthetic Network Topologies. Figure 10 compares the SIR dynamics of the real contagion network with three synthetic network models: Erdős–Rényi, Watts–Strogatz, and Barabási–Albert. The comparison shows that epidemic behavior is strongly affected by topology, but also by the initial distribution of infected nodes.

The real network exhibits the earliest epidemic peak. This behavior is not only a consequence of its topology, but also of its initial conditions: a large fraction of the nodes begin in the infected state. In contrast, the synthetic networks are initialized with fewer infected nodes, producing smoother and delayed infection curves.

Among the synthetic networks, the Erdős–Rényi and Watts–Strogatz models show faster early propagation than the Barabási–Albert network. The small-world structure of Watts–Strogatz facilitates transmission through short paths, while the Barabási–Albert topology produces a more gradual outbreak under the chosen parameterization. These results indicate that epidemic dynamics cannot be explained by the SIR parameters alone; they depend jointly on transmission probabilities, recovery rates, topology, and initial seeding.

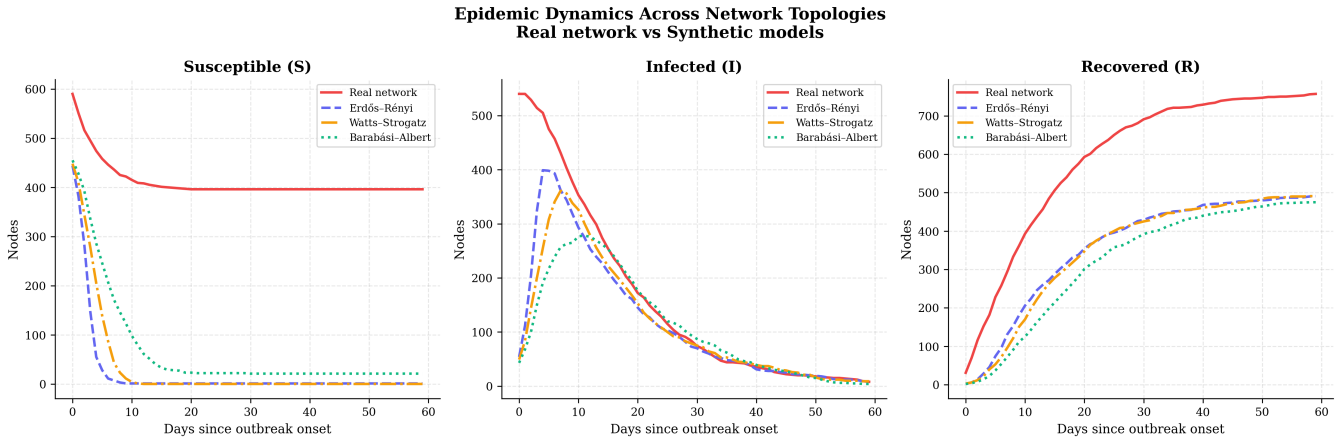


Fig. 10. Comparison of SIR dynamics across network topologies. The panels show the evolution of susceptible, infected, and recovered nodes for the real network and three synthetic models.

Discussion

The results show that epidemic propagation on graphs is determined by the interaction between epidemiological parameters, initial conditions, and network topology. In the real contagion graph, the outbreak peaks very early because a large number of nodes are initially infected. However, the sparse and fragmented structure of the network limits long-range propagation, producing many short and localized transmission chains.

Hub isolation was the most effective containment strategy. This result is consistent with the structural role of high-degree nodes: because they are connected to many neighbors, they can accelerate transmission inside local clusters. Removing these

nodes reduces the number of possible transmission routes and decreases both the infection peak and cumulative incidence. In this simulation, hub isolation reduced the peak from 540 to 435 infected nodes and reduced the final recovered count from 757 to 490.

BFS-based contact tracing achieved an intermediate reduction. Its main advantage is that it uses local information and has low computational complexity. This makes it more realistic in public health contexts where complete knowledge of the contact network is usually unavailable. Instead of requiring global centrality calculations, BFS only needs the direct contacts of known infected nodes. In the simulation, this strategy reduced the peak from 540 to 491 infected nodes.

Betweenness-based vaccination was less effective in this network. Although betweenness centrality is useful for identifying bridge nodes in connected graphs, its effectiveness decreases in highly fragmented networks. In this case, many high-betweenness nodes connect small local structures rather than large active transmission regions. Therefore, vaccinating them does not strongly alter the main epidemic trajectory. This explains why the peak only decreased from 540 to 523 infected nodes.

The comparison with synthetic networks confirms that topology changes epidemic behavior. Random, small-world, and scale-free networks produce different infection peaks and recovery patterns, even under comparable SIR parameters. This supports the central idea of the project: graph structure is not merely a visualization tool, but a mathematical component of the epidemic model. In particular, the results suggest that interventions must be selected according to the topology of the contact network. Degree-based interventions are more effective in networks where local hubs dominate transmission, while path-based interventions are more useful when large connected components and bridge nodes control global connectivity.

The contagion path analysis via Dijkstra's algorithm demonstrates the potential of shortest-path methods for retrospective epidemiological tracing. By transforming edge weights through $-\log(w_{uv})$, the algorithm identifies the sequence of contacts that maximizes the cumulative transmission likelihood between two cases. The recovered chain

$$1015 \to 1030 \to 1016$$

passes through the dominant hub, supporting the interpretation that high-degree nodes mediate some of the most probable transmission routes in the graph.

Several limitations should be considered. The dataset includes only confirmed and officially reported cases, excluding asymptomatic and undetected infections. The edge construction rule based on department and temporal proximity is an approximation and does not prove real person-to-person transmission. The model also does not include reinfection, waning immunity, time-varying behavior, vaccination campaigns over time, or viral variants. Therefore, the results should be interpreted as a computational simulation of plausible contagion dynamics rather than as a definitive reconstruction of the COVID-19 outbreak in Colombia.

Overall, the graph-based SIR framework provides a flexible computational approach for studying epidemic spread. By combining real surveillance data, graph algorithms, containment strategies, and synthetic network models, the simulator makes it possible to analyze how local connectivity and global topology shape the evolution of an epidemic.

References

[1] Instituto Nacional de Salud. *Informe de evento COVID-19, Colombia 2020*. Bogotá: INS; 2020.

[2] Barabási, A.-L., and Albert, R. Scale-free characteristics of random networks: The topology of the world-wide web.

[3] Grillo-Ardila, E. K. Mathematical model and COVID-19. *Colombia Médica*. 2020;51(2).

[4] Imai, N., et al. *Models in the COVID-19 Pandemic*. National Center for Biotechnology Information, 2024. Available at: https://www.ncbi.nlm.nih.gov/books/NBK613998/

[5] University of Michigan. *Modelling COVID-19: SIR Model*. Many Models of COVID-19. Available at: https://sites.google.com/umich.edu/manymodelsclovid19/home/the-models-of-covid-19/sir-model

[6] Wilches-Visbal, J. H., and Castillo-Pedraza, M. C. Mathematical approach of the SIR epidemiological model for COVID-19. *Revista de Salud Pública*. 2020.